Econometrics

Lecture 9b

Panel data models

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Introduction

Objectives

- Understand panel data structure
- Understand difference between fixed effects (FE) and random effects (RE) estimators

Panel data setting

Table: Life expectancy in three Caribbean countries

Country	Decade	GDP per capita	Life expectancy
Costa Rica	1960	4496.00	64.0748
Costa Rica	1970	5955.30	66.7941
Costa Rica	1980	7330.70	72.6449
Costa Rica	1990	7598.62	75.7693
Costa Rica	2000	8788.98	76.6796
Cuba	1960	3030.40	64.2087
Cuba	1970	3261.30	67.7057
Cuba	1980	4249.80	72.2980
Cuba	1990	4250.28	73.7423
Cuba	2000	3315.02	75.0463
Trinidad and Tobago	1960	10410.20	63.3020
Trinidad and Tobago	1970	12760.50	64.8271
Trinidad and Tobago	1980	17671.70	66.2813
Trinidad and Tobago	1990	14510.40	67.4495
Trinidad and Tobago	2000	16059.50	68.2033

Panel data setting

Basic panel setting:

- N units, T time periods (total $n = T \times N$)
- Each observation: y_{it} and X_{it} for $i = 1, \ldots, N$, $t = 1, \ldots, T$
- Basic model:

$$y_{it} = X_{it}\beta + \alpha_i + \varepsilon_{it}$$

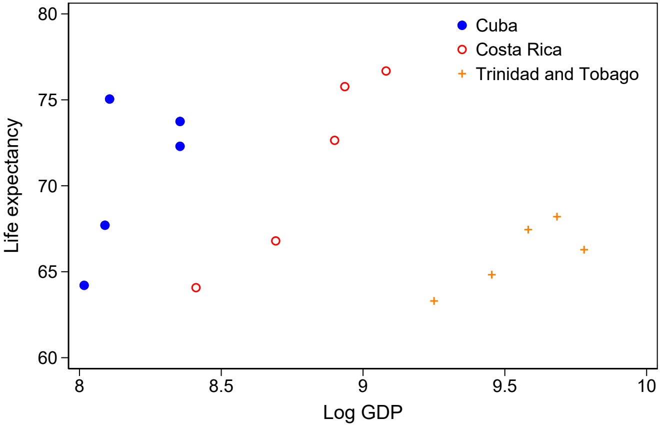
Fixed vs. random effects:

- **FE**: α_i is part of $\beta \implies$ estimate OLS
- **RE**: α_i is part of the error $u \implies$ estimate GLS

Panel data setting

Panel data setting

Does economic prosperity increase life expectancy?



Panel data setting

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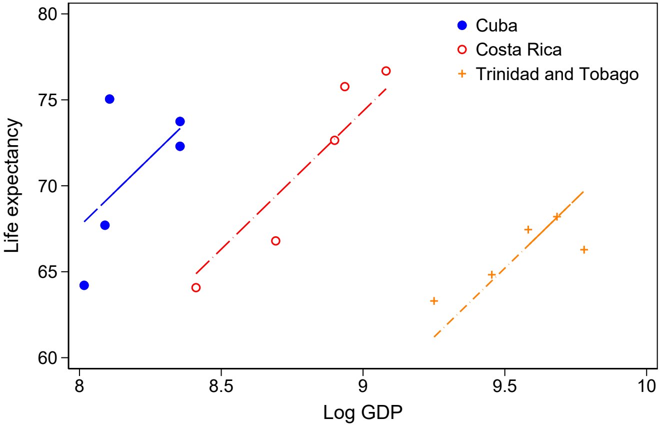

Fixed Effects (OLS)

The Fixed Effects Model

Model with unit-specific intercepts:

$$y_{it} = X_{it}\beta + \alpha_i + \epsilon_{it}, \qquad \mathbb{E}(\epsilon_{it}|X_{it}) = 0 \tag{1}$$

- ϵ_{it}: idiosyncratic error term
- α_i: **fixed effect** for unit i
- Each unit has its own intercept: $\alpha_1, \alpha_2, \ldots, \alpha_N$

Estimation

1. Via OLS including N intercepts through dummies
2. Via OLS canceling out α_i by demeaning

The Fixed Effects Model

Demeaning to eliminate α_i:

$$\bar{y}_i = \bar{X}_i\beta + \alpha_i + \bar{\epsilon}_i \tag{2}$$

$$y_{it} - \bar{y}_i = (X_{it} - \bar{X}_i)\beta + (\epsilon_{it} - \bar{\epsilon}_i) \tag{3}$$

The α_i cancel out! We can estimate β using OLS on the transformed data. Results will be equivalent.

Implications:

- Uses only **within-unit variation** over time
- Eliminates all time-invariant characteristics (observed and unobserved)
- Robust to correlation between α_i and X_{it}

The Fixed Effects Model

FE Advantages:

- Controls for **all time-invariant unobservables**
- Robust to correlation: α_i can be correlated with X_{it}
- Only requires **strict exogeneity**: $\mathbb{E}[\epsilon_{it}|X_{i1}, \ldots, X_{iT}, \alpha_i] = 0$

FE Limitations:

- Cannot estimate coefficients on time-invariant regressors
- May be **inefficient** if α_i are truly random
- Uses only within-unit variation (discards between-unit variation)

Random Effects

Random Effects Model Specification

The **random effects (RE) model** treats α_i as random:

$$y_{it} = X_{it}\beta + \alpha_i + \epsilon_{it}, \qquad \mathbb{E}(\alpha_i + \epsilon_{it}|X_{it}) = 0 \tag{4}$$

- α_i becomes part of the error term
- Remains constant for unit i across all T periods
- Example: External demand faced by industries
- α_i represents time-invariant unobserved characteristics

The composite error is: $u_{it} = \alpha_i + \epsilon_{it}$

Error Structure in Random Effects

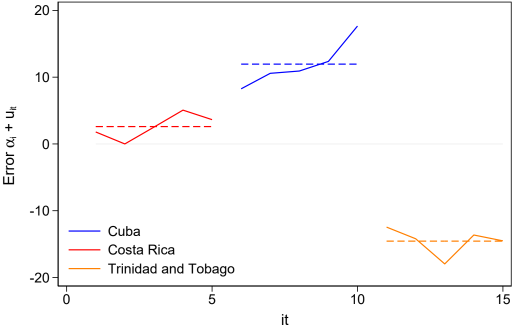

Figure: Errors with Random Effects Components

- α_i: unit-specific error component (constant over time)
- ϵ_{it}: idiosyncratic error (varies over time and units)
- Both have mean zero: $\mathbb{E}[\alpha_i] = \mathbb{E}[\epsilon_{it}] = 0$

Variance-Covariance Structure

Assumptions for the composite error $u_{it} = \alpha_i + \epsilon_{it}$:

$$\mathbb{E}[u_{it}] = 0 \tag{5}$$

$$\mathrm{var}[u_{it}] = \sigma_\alpha^2 + \sigma_\epsilon^2 \tag{6}$$

$$\mathrm{cov}[u_{it}, u_{is}] = \sigma_\alpha^2 \quad (t \neq s) \tag{7}$$

$$\mathrm{cov}[u_{it}, u_{js}] = 0 \quad (i \neq j) \tag{8}$$

For T observations within a unit:

$$\Sigma = \begin{bmatrix} \sigma_\epsilon^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_\epsilon^2 + \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\epsilon^2 + \sigma_\alpha^2 \end{bmatrix}$$

Full Variance-Covariance Matrix

For all NT observations:

$$\Omega = \begin{bmatrix} \Sigma & 0 & \dots & 0 \\ 0 & \Sigma & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \Sigma \end{bmatrix}$$

- Block diagonal structure
- Autocorrelation only due to α_i component
- Heteroskedastic: requires FGLS for efficient estimation

Feasible GLS estimator:

$$\hat{\beta}_{\mathsf{FGLS}} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}y \tag{9}$$

Estimating Variance Components

Estimate σ_ϵ^2 from within-unit variation:

$$\hat{\sigma}_\epsilon^2 = \frac{1}{N(T-K-1)} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(y_{it} - \bar{y}_i - (X_{it} - \bar{X}_i)' \hat{\beta}_{\text{OLS}} \right)^2$$

Estimate σ_α^2 using between-group estimator:

$$\bar{y}_i = \bar{X}_i \beta + \bar{u}_i \tag{10}$$

$$\hat{\beta}_B = (\bar{X}_i' \bar{X}_i)^{-1} \bar{X}_i' \bar{y} \tag{11}$$

$$\hat{\sigma}_\alpha^2 = \hat{\sigma}_B^2 - \frac{1}{T} \hat{\sigma}_\epsilon^2 \tag{12}$$

Random Effects Estimator Properties

The FGLS random effects estimator is a weighted average:

$$\hat{\beta}_{\text{RE}} = \lambda\hat{\beta}_B + (1-\lambda)\hat{\beta}_{\text{FE}} \tag{13}$$

Where λ depends on $\sigma_\alpha^2, \sigma_\epsilon^2$ and T.

- If $\sigma_\alpha^2 = 0$: $\hat{\beta}_{\text{RE}} = \hat{\beta}_{\text{FE}}$
- If $\sigma_\epsilon^2 = 0$: $\hat{\beta}_{\text{RE}} = \hat{\beta}_B$
- Uses both within and between variation
- **More efficient** than FE (if specification is correct)

Potential Problem: Endogeneity

Critical assumption: $\mathbb{E}[u_{it}|X_{it}] = 0$

If α_i is correlated with any regressor in X:

- Random effects estimator becomes **biased** (and inconsistent)
- Fixed effects estimator remains unbiased (and consistent)
- Cuba example: Low income associated with political regime and health policy

The FGLS estimator is unbiased only if:

$$\mathbb{E}[u|X] = 0 \quad \text{(orthogonality condition)}$$

Fixed or Random Effects?

- **Random Effects**: More efficient, but requires strong exogeneity
- **Fixed Effects**: Robust to correlation between α_i and X
- Most researchers use RE only if Hausman test doesn't reject

Basic decision rule:

- If α_i correlated with X: Use FE
- If α_i orthogonal to X: Use RE (more efficient)

Hausman Test

Tests null hypothesis: $\hat{\beta}_{\mathsf{RE}} = \hat{\beta}_{\mathsf{FE}}$

$$H = (\hat{\beta}_{\mathsf{FE}} - \hat{\beta}_{\mathsf{RE}})'[\Sigma(\hat{\beta}_{\mathsf{FE}} - \hat{\beta}_{\mathsf{RE}})]^{-1}(\hat{\beta}_{\mathsf{FE}} - \hat{\beta}_{\mathsf{RE}}) \sim \chi^2_k \quad (14)$$

- k: number of coefficients in β
- Large H: reject null, use Fixed Effects
- Small H: don't reject null, use Random Effects
- Similar logic to t-test: compares difference to variance

Takeaways

Takeaways

	Dependent variable: Life Expectancy			
	Pooled OLS (1)	FE dummies (2)	FE plm (3)	RE plm (4)
gdppercapita	−1.450 (2.067)	16.050*** (3.682)	16.050*** (3.682)	1.842 (3.038)
factor(country)Cuba		9.352*** (2.854)		
factor(country)Trinidad and Tobago		−17.157*** (3.239)		
Constant	82.097*** (18.329)	−70.111* (32.443)		52.974** (26.944)
Observations	15	15	15	15
R2	0.036	0.731	0.633	0.028
Note:		*p<0.1; **p<0.05; ***p<0.01		

Key issue: When is assumption A3 $\mathbb{E}(u|X) = 0$ satisfied?

Takeaways

Prefer FE over RE:

- When unobserved unit characteristics **may be correlated** with regressors
- When you want to control for **all time-invariant confounders**
- When you're only interested in **within-unit changes over time**

What Fixed Effects Captures: "How do changes in X within the same unit over time affect changes in y?"

What Fixed Effects Ignores: "Are units with higher levels of X systematically different in their levels of y?"